Grade 10 · Algebra II · Topic 2: Quadratic Functions and Equations · Lesson 2-6 — The Quadratic Formula

Classwork — Give it a go

Show all your working. Write a, b and c with their signs, and find the discriminant first. A correct answer with no working is only half an answer.

Student name :	Date : .....

Part A — The discriminant Questions 1 to 4

1. Find the discriminant and state the number and type of solutions.

$$x^2 - 7x + 10 = 0$$

2. Find the discriminant and state the number and type of solutions.

$$4x^2 - 12x + 9 = 0$$

3. Find the discriminant and state the number and type of solutions.

$$x^2 + 3x - 5 = 0$$

4. Find the discriminant and state the number and type of solutions.

$$2x^2 - 4x + 5 = 0$$

Part B — The quadratic formula Questions 5 to 7

5. Solve using the quadratic formula.

$$3x^2 + 5x - 2 = 0$$

6. Solve using the quadratic formula. Give the exact solutions.

$$x^2 - 6x + 2 = 0$$

7. Solve. Write both solutions in the form $a + bi$.

$$2x^2 - 4x + 7 = 0$$

Part C — An unknown, a context and an error Questions 8 to 10

8. Find all values of k for which the equation has exactly one real solution.

$$x^2 + kx + 25 = 0$$

9. A Jeddah Season stall has a model revenue of $R(p) = -8p^2 + 320p$ riyals at a ticket price of p riyals. Use the discriminant to decide whether a revenue of $3\,400$ SAR is possible (model values).

$$8p^2 - 320p + 3400 = 0$$

10. A student solves $x^2 - 4x - 5 = 0$ like this. Say what is wrong and give the correct solutions.

$$x = \frac{4 \pm \sqrt{-4^2 - 4(1)(-5)}}{2} = \frac{4 \pm 2}{2}$$

Part D — Choose the method Questions 11 and 12

11. Solve each equation by the method that suits it best (square roots, factoring or the quadratic formula) and name the method you chose.

(a) $x^2 - 81 = 0$ (b) $x^2 - 3x - 10 = 0$ (c) $x^2 + 3x + 5 = 0$

12. Write a quadratic equation of your own whose solutions are nonreal. Use the discriminant to prove it, then solve it.